WEEKLY PROGRESS REPORT

Mechatronic System Design (MEC483) — Course Project

PROJECT / PHASE

Smart Pick and Place Robot / Phase 2

DATE RANGE

3/3/2026 – 10/3/2026

TEAM MEMBERS

Maher Abo Abed (1087993) , Sabeeha Zainab Hasham (1089042), Basel Feras Ghunaim (1088912), Ahmad Nasser Alshehhi (1090882)

HOW ARE WE DOING THIS WEEK?

TOP WINS (MAX 3)

- Win 1: Made decisions on some major components (Computer vision, electrical actuators, displacement sensors) of the system through a systematic and objective procedure.
- Win 2: Tested popular computer vision pretrained models to check for their computability such YOLOv3 andYOLOv4 through MATLABs Computer Vision and Deep Learning Toolboxes, along with the laptop's built-in camera.
- Win 3: Explored the potential use of the AccelStepper library to simplify stepper motor control and improve motion smoothness, as it can handle acceleration and deceleration profiles, timing of step pulses, internal software-based step counting, and non-blocking control for multiple motors.

KEY BLOCKERS / RISKS (MAX 3)

- Blocker 1: No access to the university facility due to the ongoing regional situation ; access is needed to 3D print parts and begin with testing and prototyping.
- Blocker 2: Uncertainty around university campus operation status and course delivery method has hindered communications and work progress.

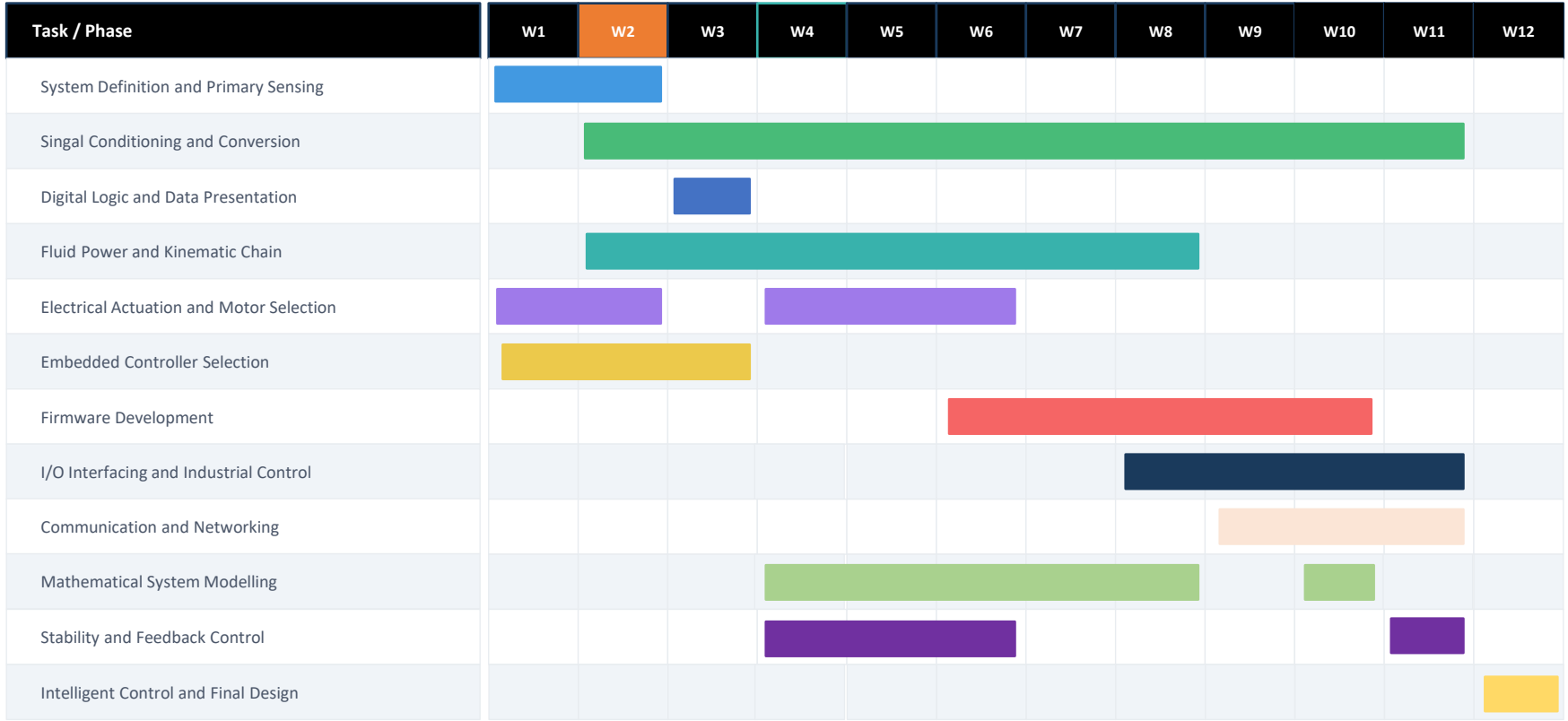
PROJECT HEALTH

Scope	Under Execution
Schedule	Potential Delays
Fabrication & Prototyping	On Halt

OPEN ISSUES

3 Active

Project Schedule (Gantt)



MILESTONES & DELIVERABLES

Milestone / Deliverable	Due Date	Owner	Status / Evidence
Sensor System Definition (Week 1 Report)	–	Member 3 & Member 4	COMPLETED / PDF Documentation
Motor Selection & Analysis (PCC Method)	–	Member 2	COMPLETED / Decision Matrix
Encoder & Position Strategy Evaluation	–	Member 1 & Member 3	COMPLETED / Technical Review
Vacuum Suction System Design	–	Member 3	COMPLETED / Force Calculation
Pump Selection Finalization	–	Member 2	IN PROGRESS
Hybrid End Effector Concept (Gripper + Suction)	–	Member 2 & Member 3	IN PROGRESS
Sensor Integration & System Architecture	–	Member 4 & Team	IN PROGRESS
Motion Control Implementation (Stepper Control Code)	–	Member 3	PENDING
CAD Model of Prototype	–	Member 1 & Team	IN PROGRESS
Mechanical Prototype Assembly	–	Team	NOT STARTED
System Integration Testing	–	Team	NOT STARTED
Final Validation & Demonstration	–	All Members	NOT STARTED

ISSUES, RISKS, AND ACTION ITEMS

TOP ISSUES (BLOCKING PROGRESS)

- Being limited to solely software-based work, ie: research/computer vision can slow down performance – not having easy access to facilities to make the model, ie: 3D printer.
- Due to current circumstances, teammates cannot meet in person to discuss project planning, this has resulted in Microsoft Teams meetings occurring 1-2 times a week. This can be limiting as only one teammate has access to the objects that need to be picked up by the arm.

MITIGATIONS / DECISIONS NEEDED

- Focus on aspects that can be conducted without the need for printing the model – design/computer vision/research
- More comprehensive meetings are required to mitigate this issue, planning in advance and strong delegation of tasks needs to be implemented.

ACTION ITEMS (NEXT 7 DAYS)

ID	ACTION ITEM DESCRIPTION	MEMBER	WEEK DUE
A1	Design a useable end effector and complete the pump selection	Sabeeha	Week 4
A2	Exploring computer vision models and develop a model that can be used to test on	Maher	Week 4
A3	Investigate motion control strategy, torque estimations, suction system and gripper	Basel	Week 4
A4	Finalise sensor selection and integration in system	Ahmad	Week 4

INDIVIDUAL UPDATES

WEEKLY PROGRESS & NEXT STEPS

MEMBER 1

Maher Abo Abed

MEMBER 2

Sabeeha Zainab Hasham

MEMBER 3

Basel Feras Ghunaim

MEMBER 4

Ahmad Nasser Alshehhi

MEMBER 1: Maher Abo Abed

ACCOMPLISHED (SINCE LAST REPORT)

- Tested computer vision models such as YOLOv3, YOLOv4 through MATLABs Toolboxes.
- Explored the need and the types of angular displacement sensors that can be used and their sub-types.
- Identified optical incremental encoder as feasible option.

IN PROGRESS

- Develop a model that can be used for testing and familiarization of the different sub-systems and components..
- Exploring various computer vision models that can be used, as an alternative to training an AI model from scratch.

NEXT WEEK PLAN

- Finalize selection of most components through an objective evaluation.
- Make significant progress on a CAD model for a testing prototype. .

EVIDENCE

HELP NEEDED / QUESTIONS

- Is access to university equipment and facility possible throughout the break?
- Can a reliable system be constructed without the use of angular displacement sensors such as encoders?

MEMBER 2: Sabeeha Hasham

ACCOMPLISHED (SINCE LAST REPORT)

- Comprehensive motor selection and analysis using PCC method, 3 options investigated in 6 categories
  – weighted decision matrices
- Defined the criteria required for the pump selection and initiated an analysis
  Started looking at potential end effectors – potentially to use a rack and pinion and servo motor

IN PROGRESS

- Completion of pump selection - in progress
- Completion of the end effector design to be completed

NEXT WEEK PLAN

- Research and on computer vision – potential trial runs.

EVIDENCE (SHOW, DON'T TELL)

Progress report consisting of already conducted research is attached to this slide.

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HELP NEEDED / QUESTIONS

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MEMBER 3: Basel Feras Ghunaim

ACCOMPLISHED (SINCE LAST REPORT)

- Established a position sensing system using homing limit switches and stepper motor step counting.
- Considered the necessity of a velocity sensor and ensured that velocity could be calculated in software.
- Design and documentation of the vacuum suction system to handle CDs, calculation of holding force and component selection.
- Evaluated encoder feedback as optional improvement to better positioning reliability.
- Added gear and belt transmission ratios to calculations of joint positions.

Evidence:
stteper motor and encoder.docx
Mechatronics System Design week one.docx
Vacuum Suction System.docx

IN PROGRESS

- Last comparison of motion control implementation software (AccelStepper library vs home written step pulse generating).
- Determining the need or not of encoder feedback, according to anticipated mechanical load and possible step loss.
- Optimizing the incorporation of the hybrid end-effector design (mechanical fingers + suction cup).
- Revising the effect of gear backlash and belt elasticity on positioning error.

NEXT WEEK PLAN

- Make decision on motion control strategy final and start implementation.
- If encoder feedback is used, begin basic planning for PID control.
- Perform torque estimations based on transmission ratios and loading circumstances.
- Begin mechanical layout planning of the installation of the suction system in the gripper assembly.

HELP NEEDED / QUESTIONS

- Clarification on whether rotary encoders are strictly required for this project, or if open-loop stepper control with homing limit switches is sufficient.
- Confirmation on the acceptable level of positioning accuracy expected for grading purposes.
- Guidance on whether encoder-based closed-loop feedback is recommended given the current mechanical design (gear and belt transmission).

MEMBER 4: Ahmed Alshehhi

ACCOMPLISHED (SINCE LAST REPORT)

- Researched and identified the required sensors for the pick-and-place robot system.
- Evaluated different sensor options and prepared a list of recommended sensors with their purposes and approximate costs.

Evidence : Sensor specification list and preliminary components comparison for decision matrix.

IN PROGRESS

- Assisting in defining the sensor integration architecture for the robotic system.

NEXT WEEK PLAN

- Finalize the sensors selection based on the decision matrix results.
- Assist in integrating the selected sensors into the initial system design and CAD model.

HELP NEEDED / QUESTIONS

- Clarification on the final testing parameters for the robotic systems.

WRAP-UP: WHAT WE NEED FROM THE INSTRUCTOR / STAKEHOLDERS

DECISIONS REQUESTED

- Access to university laboratories for 3D printing the required components and models.

ACTION ITEMS FOLLOW-UP

- Begin working on the initial CAD design draft.
- Design, build, and test a small-scale prototype before developing the final system.

QUESTIONS / FEEDBACK

- When will we have access to the university labs for fabrication and testing?
- What are the testing parameters and evaluation criteria for the prototype?

APPENDIX

- Sensor specification sheets.
- MATLAB Computer Vision Code